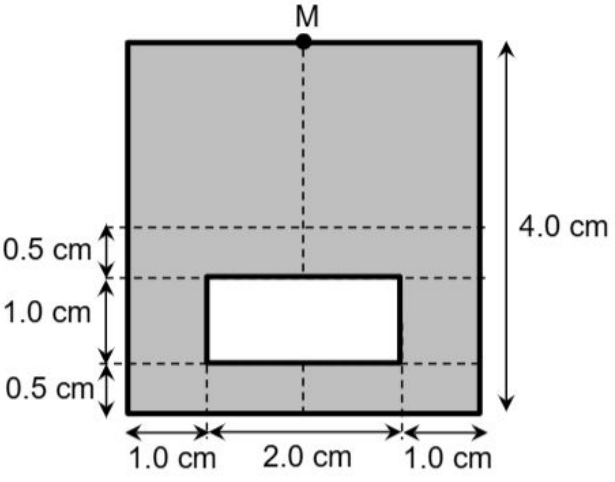
3 A uniform sheet of metal of height 4.0 cm and width 4.0 cm has a rectangular hole, measuring 2.0 cm by 1.0 cm, cut in it as shown below.

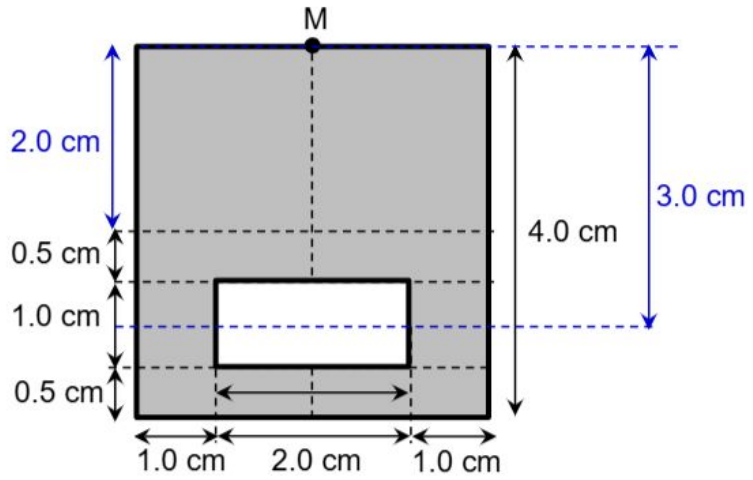


Point M is the mid-point of the top of the sheet.

What is the distance from point M of the centre of gravity of the sheet of metal?

	A	1.7 cm	**B**	1.9 cm	**C**	2.1 cm	**D**	2.3 cm

L3 **Answer: B**



Let density of sheet to be ρ and thickness to be t.
Let W_S be the weight of the original sheet **without** hole, and,
W_H be the weight of the part cut out to form the hole.

Taking the pivot at point M,
Sum of moments for weight of individual components about M
= Moment of W_S about M – Moment of W_H about M
= $\rho A_S t g (2.0) - \rho A_H t g (3.0)$
(*The second term is subtracted from the first as the hole is cut out from the sheet.)
= $\rho (4.0)(4.0) t g (2.0) - \rho (1.0)(2.0) t g (3.0)$
= $26 \rho t g$

Let the distance from M of the effective centre of gravity of the sheet with hole be y.
Moment of the effective weight of sheet with hole about M
= $(W_S - W_H) y$
= $[\rho (4.0)(4.0) t g - \rho (1.0)(2.0) t g] y$
= $14 \rho t g y$

Therefore,
$26 \rho t g = 14 \rho t g y$
$y = 1.857 =$ **1.9 cm**